The empirical recurrence for OEIS A210176, proved by transfer matrix

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Abstract

OEIS A210176 counts arrays of width 2 whose every 2×2 subblock holds a prescribed number of DISTINCT values, which use ALL of the 4 available letters, and which are counted up to renaming. The entry carries an empirical recurrence of order 4 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition counts distinct values and so survives renaming, which lets the count of patterns be recovered from counts over fixed alphabets by inverting a triangular system of falling factorials; the words "containing all values" change which row of that inverse is used and nothing else. Each fixed-alphabet count is a walk count on $S = 30$ states in all.

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1 The conjecture

OEIS A210176 is "Number of (n+1) X 2 0..3 arrays containing all values 0..3 with every 2 X 2 subblock having three or four distinct values, and new values 0..3 introduced in row major order.". It has offset 1 and begins

$$1, 51, 744, 9042, 103752, 1165620, 12988416, 144259560, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 14*a(n-1) - 26*a(n-2) - 68*a(n-3) - 24*a(n-4).

As of the "Last modified" line on the live entry (Jul 15 2018 06:53:23, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The array has 2 cells across and entries in $\{0, 1, \ldots, 3\}$. In every 2×2 window of it, the number of DISTINCT entries must be one of $3, 4$. Two further clauses come with it:

- "new values 0..3 introduced in row major order" — reading the cells in row major order, a letter may appear for the first time only when every smaller letter already has, so the array counted is the canonical representative of its EQUALITY PATTERN;
- "containing all values 0..3" — every one of the 4 letters must actually occur.

3 From patterns to fixed alphabets

Write N_j for the number of admissible patterns using exactly j letters and L_i for the number of admissible arrays over an alphabet of i letters. An array over i letters is a pattern together with an injection of its letters into the alphabet, so

$$L_i \;=\; \sum_j N_j\, i(i-1)\cdots(i-j+1) \;=\; \sum_j F_{ij} N_j,$$

a triangular system with unit diagonal, hence invertible over $\mathbf{Q}$.

Here is the whole difference between this entry and its neighbours. Without the words "containing all values" the quantity counted is $N_1 + \cdots + N_4$, and the weights are the row vector $\mathbf{1}^\top F^{-1}$. With them, the pattern must use every letter, so the quantity counted is N_4 alone and the weights are the single row $e_4^\top F^{-1}$ of the same inverse:

$$a \;=\; N_4 \;=\; \sum_{i=1}^{4} \left(F^{-1}\right)_{4i} L_i.$$

The step is legitimate only if the condition survives renaming the alphabet, and it does: the condition counts how many DISTINCT values a window holds, and whether two cells agree is untouched by any renaming. No clause compares letters by size or does arithmetic on them.

4 Each L_i is a walk count

A 2×2 window lies in 2 consecutive array rows, so the window of the last 1 rows is a state and a step appends a row, settling every subblock it completes. Assembling the pieces for $i = 1, \ldots, 4$ as a disjoint union, with ι carrying the weights above cleared of denominators and $\tau = \mathbf{1}$,

$$a(n) \;=\; \iota^\top M^{\,j} \tau, \qquad S = 30.$$

The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

5 The proof

Lemma 1. *Let $q(t) = t^4 - \sum_i c_i t^{4-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^\top M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+4} - \sum_i c_i M^{j+4-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j \;=\; \iota^\top M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 14\, a(n-1) - 26\, a(n-2) - 68\, a(n-3) - 24\, a(n-4)$$

for every $n > 4$.

Proof. The vector $w = q(M)\tau$ was formed by 4 matrix–vector products in exact integer arithmetic and $\iota^\top M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 4 + 1$ onwards and the run continued until the working vector was identically zero. The rational weights are carried as one common denominator throughout, so no division is performed until the final term. □

6 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: enumerate every array over $\{0, \ldots, 3\}$, keep those that are canonical, that use every letter, and whose every subblock passes, and count them. Neither the transfer matrix nor the falling-factorial inversion is involved, and the published terms came back — including the initial zeros, which are a sharp test of the "containing all values" clause: a small array cannot show 4 letters at all.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A210176.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7; Stirling numbers and falling factorials, Section 1.9.)

[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.